

Can Marcus Trust His Flight?

Meet Marcus. He flies **80 times a year** — and has missed **3 connections this quarter alone**. He's done guessing which airline to trust.

"I am done guessing which airline to trust."

Datafonos · Daniel Guilabert · Alberto Rojas · Miguel Mercade · BTS 2023



Made with GAMMA



That's Not Bad Luck. That's the System.

256.8M

Hours lost to delays in 2023 — across **6.7 million flights** analyzed nationwide.

1 in 5

Flights arrive late

6.6 min

Average delay per flight

6.7M

Flights analyzed

79.4%

On-time rate nationwide



Marcus Goes Detective

"I always book Friday afternoon. That is literally the worst slot of the week."

● Worst Airports

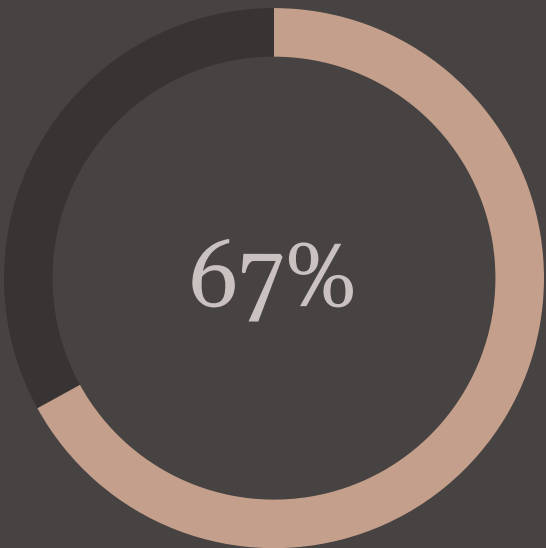
Orlando MCO · **15.8 min** avg delay
Newark EWR · **15.1 min**
Miami MIA · **14.4 min**

✓ Best Slot to Fly

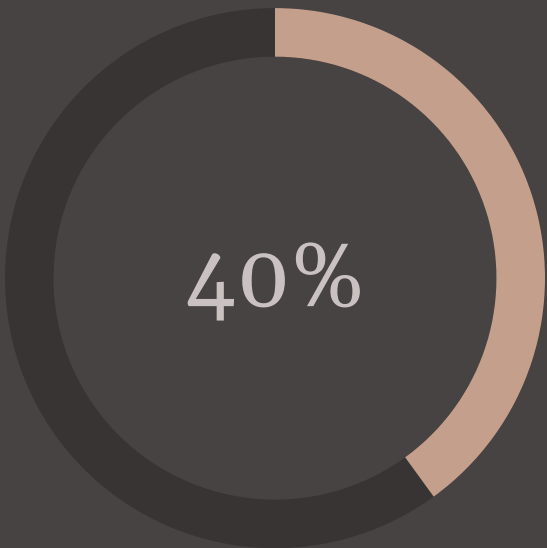
Tue/Thu before 06:00 → arrives early
Fri/Sun 19–22h → up to **19 min** delay

The Verdict Is In

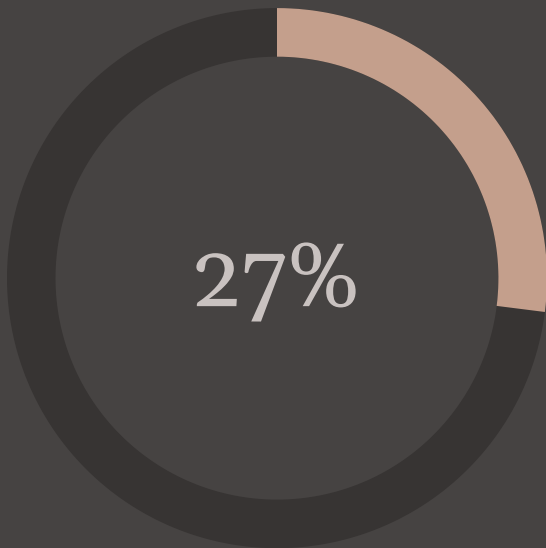
67% of delays are the airline's fault. Weather is NOT the main villain — it accounts for just 11% of disruptions.



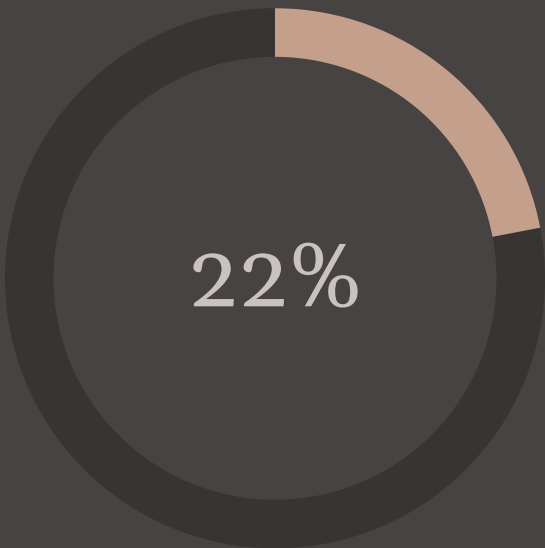
Airline's Fault



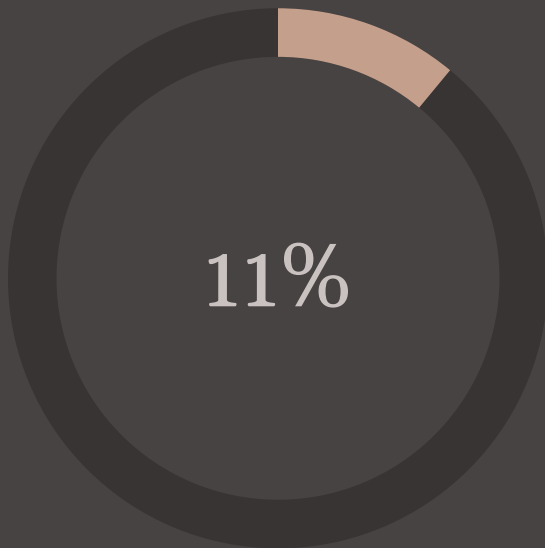
Late Incoming Aircraft



Airline Operations



Air Traffic Control



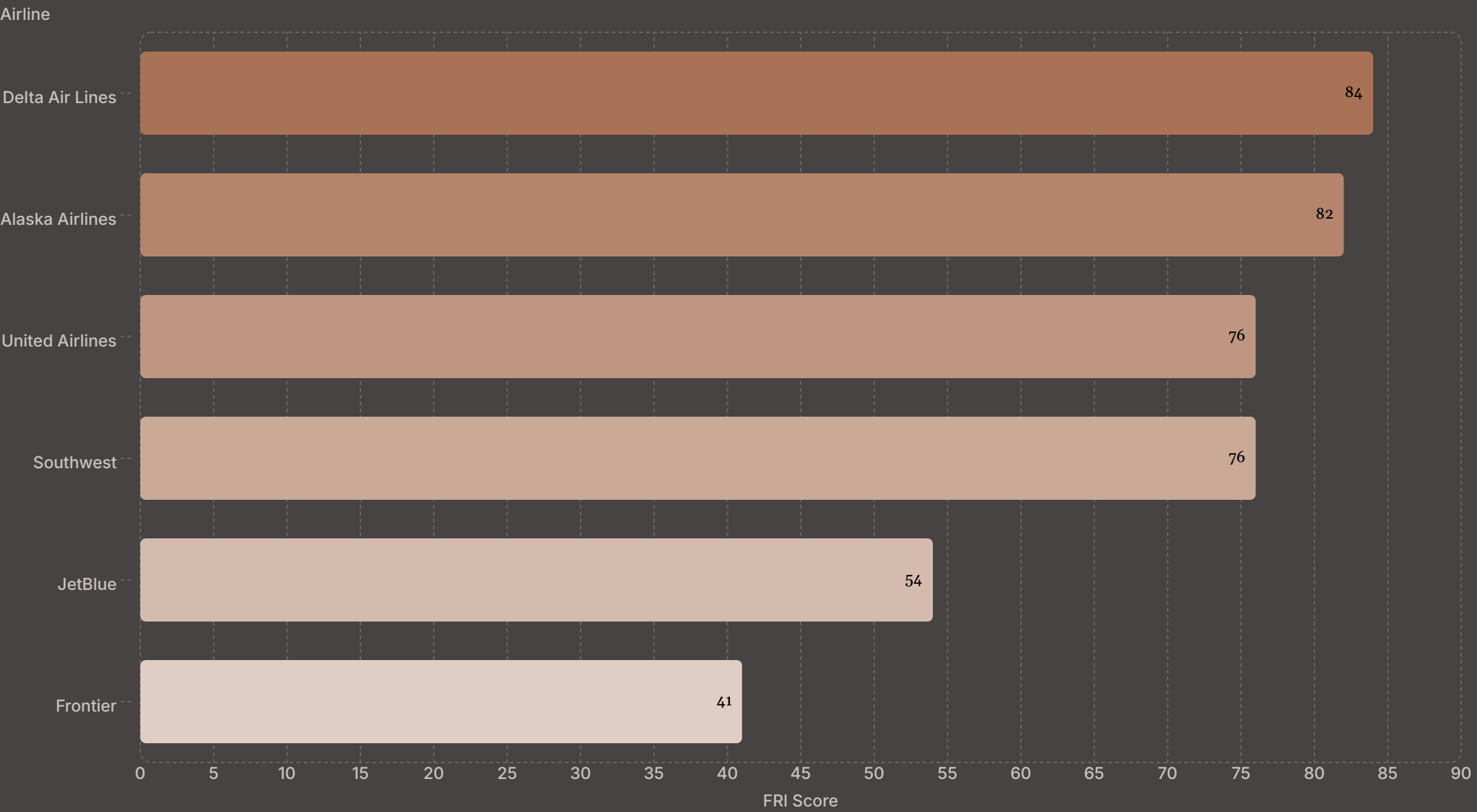
Weather

"It IS the airline. I can fix this."

Marcus Enters the Lab

Using BTS 2023 data, the team built a **Flight Reliability Index (FRI)** — a composite score weighting on-time performance, average delay, and cancellation rate.

$$\text{FRI} = \text{On-time} \times 50\% + \text{Delay} \times 30\% + \text{Cancellation} \times 20\%$$



43 points separate Delta from Frontier — invisible from average delay data alone.

"I have been flying JetBlue to save \$40. That was a mistake."

The Domino Effect

30 minutes late departing = 28 minutes late arriving. Airlines don't recover time in the air — confirmed across 6.7 million flights. A late plane stays late.

Severe Delay Leaderboard (*flights over 1 hour late*)

JetBlue

13.8%

1 in 7 flights

Frontier

12.3%

1 in 8 flights

Delta

5.5%

Best major carrier

Alaska

4.3%

Most consistent



So Marcus Built a Tool.

Four modules. One mission: **book with confidence, not luck.**



Airport Congestion Map

Identify high-risk departure hubs before you book



Hour × Weekday Heatmap

Pick the safest departure window by day and hour



FRI Carrier Ranking

Compare airlines on reliability, not price alone



Severe Delay Analysis

Flag carriers with the highest rate of 1+ hour disruptions



Marcus Knows What to Do Now.

Delta. Tuesday morning. Never Friday evening.

✓ Book Delta or Alaska FRI score 82–84. The most reliable major carriers in the dataset.	✓ Fly Tue/Thu before 9am The best on-time window of the entire week — consistently.
☒ Avoid JetBlue & Frontier For tight connections, a 13–14% severe delay rate is too risky.	☒ Never Sun/Fri evenings Up to 19 min average delay. The system is already exhausted by then.